

Fundamental Trigonometric Identities

ANSWER KEY



Using the Pythagorean identities

- 1 Given that $\sin \theta = \frac{3}{5}$ and θ lies in Quadrant II, find:
- a** $\cos \theta = -\frac{4}{5}$ **b** $\tan \theta = -\frac{3}{4}$ **c** $\sec \theta = -\frac{5}{4}$ **d** $\csc \theta = \frac{5}{3}$
- 2 Given that $\tan \theta = 2$ and θ lies in Quadrant III, find $\sin \theta$ and $\cos \theta$ in exact simplified form.
 $\sec^2 \theta = 1 + \tan^2 \theta = 5$, and in QIII $\cos \theta < 0$, so $\cos \theta = -\frac{\sqrt{5}}{5}$ and $\sin \theta = \tan \theta \cos \theta = -\frac{2\sqrt{5}}{5}$.
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Simplifying and verifying

- 3 Simplify each expression to a single trigonometric function:
- a** $\frac{1 - \cos^2 x}{\sin x} = \frac{\sin^2 x}{\sin x} = \sin x$
- b** $\sec x \cot x = \frac{1}{\cos x} \cdot \frac{\cos x}{\sin x} = \csc x$
- c** $\tan x \csc x = \frac{\sin x}{\cos x} \cdot \frac{1}{\sin x} = \sec x$
- 4 Verify the identity $(1 + \tan^2 x) \cos^2 x = 1$.
 $(1 + \tan^2 x) \cos^2 x = \sec^2 x \cos^2 x = \frac{1}{\cos^2 x} \cos^2 x = 1$.
- 5 Verify the identity $\frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}$.
Cross-multiplying view: $\sin^2 x = (1 + \cos x)(1 - \cos x) = 1 - \cos^2 x$, which is the Pythagorean identity.
Equivalently, multiply the left side by $\frac{1 - \cos x}{1 - \cos x}$ to obtain $\frac{\sin x(1 - \cos x)}{1 - \cos^2 x} = \frac{\sin x(1 - \cos x)}{\sin^2 x} = \frac{1 - \cos x}{\sin x}$.
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Quotient and reciprocal identities

- 6 Rewrite each expression using only $\sin x$ and $\cos x$, then simplify:
- a** $\cot x \sin x = \frac{\cos x}{\sin x} \cdot \sin x = \cos x$
- b** $\frac{\sec x}{\csc x} = \frac{1/\cos x}{1/\sin x} = \frac{\sin x}{\cos x} = \tan x$
- c** $1 - (\sin^2 x \csc^2 x) \sin^2 x = 1 - \sin^2 x = \cos^2 x$ (since $\csc^2 x \sin^2 x = 1$)
- 7 Simplify $\frac{\cos x}{1 - \sin x} - \tan x$ to a single term.
Combine over $\cos x(1 - \sin x)$: $\frac{\cos^2 x - \sin x(1 - \sin x)}{\cos x(1 - \sin x)} = \frac{\cos^2 x + \sin^2 x - \sin x}{\cos x(1 - \sin x)} = \frac{1 - \sin x}{\cos x(1 - \sin x)} = \sec x$.
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More identity verifications

- 8** Verify the identity $\sec^2 x - \tan^2 x = 1$ using the Pythagorean identity for tangent and secant.
The identity $1 + \tan^2 x = \sec^2 x$ (dividing $\sin^2 x + \cos^2 x = 1$ by $\cos^2 x$) rearranges directly to $\sec^2 x - \tan^2 x = 1$.
- 9** Verify the identity $\frac{1}{1 - \cos x} + \frac{1}{1 + \cos x} = 2\csc^2 x$.
Combine over a common denominator: $\frac{(1 + \cos x) + (1 - \cos x)}{(1 - \cos x)(1 + \cos x)} = \frac{2}{1 - \cos^2 x} = \frac{2}{\sin^2 x} = 2\csc^2 x$.
- 10** Show that $(\sin x + \cos x)^2 = 1 + 2\sin x \cos x$.
Expand:
 $(\sin x + \cos x)^2 = \sin^2 x + 2\sin x \cos x + \cos^2 x = (\sin^2 x + \cos^2 x) + 2\sin x \cos x = 1 + 2\sin x \cos x$.
- 11** Given that $\cos \theta = -\frac{8}{17}$ and θ lies in Quadrant III, find $\sin \theta$, $\tan \theta$, and $\cot \theta$ in exact simplified form.
 $\sin^2 \theta = 1 - \frac{64}{289} = \frac{225}{289}$, and in QIII $\sin \theta < 0$: $\sin \theta = -\frac{15}{17}$. $\tan \theta = \frac{-15/17}{-8/17} = \frac{15}{8}$. $\cot \theta = \frac{8}{15}$.

Identities with sums and products

- 12** Verify the identity $\sin x \cos x (\tan x + \cot x) = 1$.
 $\sin x \cos x \left(\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} \right) = \sin^2 x + \cos^2 x = 1$.
- 13** Simplify $(\sec x - \tan x)(\sec x + \tan x)$ to a constant.
This is a difference of squares: $\sec^2 x - \tan^2 x = 1$ (a Pythagorean identity).